Let $SZ = 2a$. Then S is $(a, 0)$ and the equation of the directrix is $x + a = 0$.

Let $P(x, y)$ be the moving point in the locus that yield a parabola. Draw PM perpendicular to the directrix. By definition, $e = \frac{SP}{PM} = 1$. So, $SP^2 = PM^2$.

Then, $(x-a)^2 + y^2 = (x+a)^2$. On simplifying, we get $y^2 = 4ax$ which is the equation of the **parabola in the standard form**.

The other standard forms of parabola are $y^2 = -4ax, x^2 = 4ay$, and $x^2 = -4ay$.

Definition 5.3

- The line perpendicular to the directrix and passing through the focus is known as the **Axis** of the parabola.
- The intersection point of the axis with the curve is called **vertex** of the parabola
- Any chord of the parabola, through its focus is called **focal chord** of the parabola
- The length of the focal chord perpendicular to the axis is called **latus rectum** of the parabola

Example 5.14

Find the length of Latus rectum of the parabola $y^2 = 4ax$.

Solution

Equation of the parabola is $y^2 = 4ax$.

Latus rectum LL' passes through the focus $(a, 0)$. Refer (Fig. 5.18)

Hence the point L is (a, y_1).

Therefore $y_1^2 = 4a^2$.

Hence $y_1 = \pm 2a$.

The end points of latus rectum are $(a, 2a)$ and $(a, -2a)$.

Therefore length of the latus rectum $LL' = 4a$.

Note

The standard form of the parabola $y^2 = 4ax$ has for its vertex $(0, 0)$, axis as x-axis, focus as $(a, 0)$. The parabola $y^2 = 4ax$ lies completely on the non-negative side of the x-axis. Replacing y by $-y$ in $y^2 = 4ax$, the equation remains the same. so the parabola $y^2 = 4ax$ is symmetric about x-axis; that is, x-axis is the axis and symmetry of $y^2 = 4ax$

(ii) Parabolas with vertex at (h, k)

When the vertex is (h, k) and the axis of symmetry is parallel to x-axis, the equation of the parabola is either $(y-k)^2 = 4a(x-h)$ or $(y-k)^2 = -4a(x-h)$ (Fig. 5.19, 5.20).

When the vertex is (h, k) and the axis of symmetry is parallel to y-axis, the equation of the parabola is either $(x-h)^2 = 4a(y-k)$ or $(x-h)^2 = -4a(y-k)$ (Fig. 5.21, 5.22).

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